

SCDS TileDB-SOMA Data — New User Guide

A guide to the single-cell data stored as [TileDB-SOMA](#) objects in `s3://rancho-scds-collections/`: what's there, how it's organized, and how to start querying it.

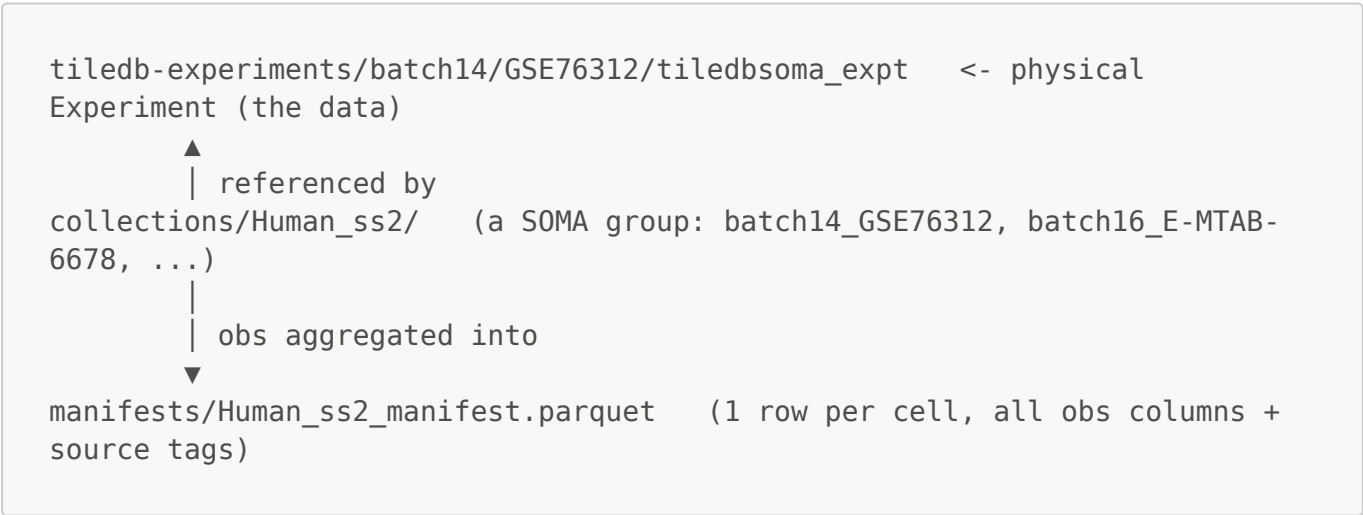
Metrics in this guide were captured on **2026-07-24**.

1. What's in the bucket

Everything lives under `s3://rancho-scds-collections/`, organized into three prefixes:

Prefix	What it holds
<code>tiledb-experiments/</code>	The actual data. One TileDB-SOMA Experiment per dataset (cells × genes: counts, normalized data, cell/gene annotations, embeddings). Grouped into ingestion <code>batch14/ ... batch19/</code> .
<code>collections/</code>	TileDB-SOMA Collections — lightweight groups that <i>reference</i> a curated set of experiments. Four collections split by organism and platform: <code>Human_10x</code> , <code>Human_ss2</code> , <code>Mouse_10x</code> , <code>Mouse_ss2</code> .
<code>manifests/</code>	One Parquet manifest per collection — an aggregated index of every cell's <code>obs</code> metadata across all experiments in that collection, for fast metadata queries without opening each experiment.

How the pieces relate



- **Experiments** are the source of truth and hold all expression matrices.
- **Collections** are just groups of references (a few KB each) — opening a collection does not copy data; it points at experiments under `tiledb-experiments/`.
- **Manifests** are a flat metadata cache built from the collections. Use them to explore and filter cell metadata, then open the underlying experiments to pull expression data.

Not every experiment is wired into a collection: there are **189** experiments in `tiledb-experiments/` but only **159** are currently included across the four collections. The other **30**

experiments are not in any collection and are therefore **not reflected in the metrics below** (which are derived from the collection manifests). These are **intentionally excluded**: they started from **author-provided count matrices** rather than being reprocessed from raw FASTQ through the standard SCDS pipeline, so their included genes may not be completely consistent with the pipeline-processed datasets in the collections. Those 30 hold **~10.0M** cells — almost as many as the collections themselves — including several large datasets:

Excluded store	Cells
batch19/CIMA	5,846,268
batch19/dat-ztfn3cc	851,922
batch19/GSE274546	644,098
batch16/LymphoMAP	564,146
batch15/GSE269981	390,756
... 25 more	~1.7M

So the full physical footprint is **~24.3M cells across 189 experiments**; the collection metrics below cover the **14.27M cells in the 159 experiments starting from FASTQ**.

2. Summary metrics

As of 2026-07-24. These counts cover only the **159 experiments included in the four collections** (they're computed from the manifests). They exclude the 30 experiments described above that start from author provided counts (~10.0M additional cells) — see [What's in the bucket](#).

Collection	Experiments	Cells	Genes	Median cells/expt	Max cells/expt
Human_10x	104	11,827,751	36,601	63,482	794,782
Human_ss2	7	14,853	36,522	1,102	7,128
Mouse_10x	46	2,420,528	32,285	34,276	627,495
Mouse_ss2	2	3,414	31,992	1,707	3,050
Total	159	14,266,546	—	—	—

Storage footprint (in [tiledb-experiments/](#), the physical data):

Metric	Value
Total data size	~225 GiB (241.6 GB)
S3 objects	185,228
Experiments	189 (batches 14-19)

Metric	Value
Manifest files	4 (~1.6 GB total)

Collections themselves are negligible in size (reference groups, a few KB each).

3. Getting set up

You need the `tiledsoma` conda environment (includes `tiledsoma`, `pandas`, `pyarrow`, `s3fs`, `scanpy`) and AWS credentials with read access to the bucket.

```
conda env create -f tiledbsoma_environment.yml # first time only
conda activate tiledbsoma
```

Quick check that everything works:

```
import tiledbsoma as soma
import pandas as pd

# Read a collection's manifest straight from S3
manifest = pd.read_parquet(
    "s3://rancho-scds-collections/manifests/Human_ss2_manifest.parquet"
)
print(manifest.shape) # (cells, columns)
```

4. Common tasks — where to look

Worked, runnable examples for all of the below are in `tiledbsoma_examples.ipynb`:

I want to...	Approach
Explore cell metadata across a whole collection	Load its manifest with <code>pd.read_parquet(...)</code> and use normal pandas (<code>value_counts</code> , filtering, <code>groupby</code>). See the Manifests section.
Open one experiment and inspect <code>obs</code> / <code>var</code> / <code>X</code>	<code>soma.Experiment.open(<expt_uri>)</code> . See the Experiments section.
Query cells by annotation (e.g. cell type) and pull expression	<code>ExperimentAxisQuery</code> with an <code>obs_query</code> value filter → <code>to_anndata()</code> . See Slicing and Filtering .
Find which experiments contain a cell type across a collection	<code>query_cell_type_in_collection(...)</code> in the Collections section.

I want to...	Approach
Find which experiments express a gene	<code>find_experiments_expressing_gene(...)</code> — uses a <code>var_query</code> (gene lives in <code>X</code> , not <code>obs</code>). See the Collections section.

Key identifiers

- **Experiment URI:** `s3://rancho-scds-collections/tiledb-experiments/<batch>/<dataset>/tiledbsoma_expt`
- **Collection URI:** `s3://rancho-scds-collections/collections/<Organism_Platform>`
- **Manifest URI:** `s3://rancho-scds-collections/manifests/<Organism_Platform>_manifest.parquet`
- **Gene identifiers** (in `var`): `var_id / gene_symbols = symbol` (e.g. `CD8A`); `gene_ids =` Ensembl ID.
- **Cell-type prediction** (in `obs`): `popv_prediction_ontology_name` (consensus popV call).